First Order Logic (FOL)

domain / Universe: the set where the elements come from

Predicate / function $\quad P(x)$: the staderent is true for variable x

$\forall$ for all $\qquad \exists$ for some (there exists one)

$U = \{x_1, x_2, \cdots, x_n\}$

$\forall x\, P(x) \equiv P(x_1) \wedge P(x_2) \wedge \cdots \wedge P(x_n)$

↳ $P(\cdot)$ holds for all elements of U

$\exists x\, P(x) \equiv P(x_1) \vee P(x_2) \vee \cdots \vee P(x_n)$

↳ $P(\cdot)$ hods for some element of U

There is an element of U for which $P(\cdot)$ holds exmp: Some babies are cute

$U = \{\text{all babies}\}$

$C(x)$: x is cute

$\exists x \, C(x)$

all babies are cute

$\forall x \, C(x)$

an alternative solution

$U = \{\text{all people}\}$

$C(x)$: x is cute

$B(x)$: x is a baby

$\exists x \, (B(x) \land C(x))$

all babies are cute

$\forall x \, (B(x) \land C(x))$: everyone is a cute baby

$\forall x \, (B(x) \Rightarrow C(x))$

No baby is cute $\boxed{\sim \forall x \, (B(x) \Rightarrow C(x))}$ ✗

$\forall x \, (B(x) \Rightarrow \sim C(x))$

some babies are not cute

De Morgan: $\sim \forall x\, P(x) \equiv \exists x \sim P(x)$

$\sim \exists x\, P(x) \equiv \forall x \sim P(x)$

short proof

$U = \{x_1, x_2, \ldots, x_n\}$

D.M from

$\sim(p \wedge q) \equiv \sim p \vee \sim q$

$$\sim \forall x\, P(x) \equiv \sim (P(x_1) \wedge P(x_2) \wedge \ldots \wedge P(x_n))$$

$$\equiv \sim P(x_1) \vee \sim P(x_2) \vee \ldots \vee \sim P(x_n)$$

$$\equiv \exists x \sim P(x)$$

$$\sim \forall x\, (B(x) \Rightarrow C(x)) \equiv \exists x \sim (B(x) \Rightarrow C(x)) \quad \text{D.M.}$$

$$\equiv \exists x \sim (\sim B(x) \vee C(x)) \quad \text{Implication}$$

$$\equiv \exists x (\sim\sim B(x) \wedge \sim C(x)) \quad \text{D.M.}$$

$$\equiv \exists x (B(x) \wedge \sim C(x))$$

Some babies are not cute

Inference Rules for FOL

$$\frac{\forall x\, P(x)}{P(c)}$$

UI (Universal Instantiation)

UE (Universal Elimination)

c : variable (refers to every element of universe)

c : constant (" " one element of universe)

$$\frac{P(c)}{\forall x\, P(x)}$$

c is variable

Universal Generalization (UG)

$$\frac{\exists x\, P(x)}{P(c)}$$

Existential Instantiation (EI)

Existential Elimination (EE)

c is a constant

$$\frac{P(c)}{\exists x\, P(x)}$$

Existential Generalization (EG)

c is constant or variable

Prove that $\underbrace{\exists x\,(\alpha(x) \wedge \beta(x))}_{\text{Hyp.}} \Rightarrow \underbrace{\exists x\, \alpha(x) \wedge \exists x\, \beta(x)}_{\text{Conc.}}$

1. $\exists x\,(\alpha(x) \wedge \beta(x))$ Hyp.
2. $\alpha(c) \wedge \beta(c)$ EI ① c: constant
3. $\alpha(c)$ Simp ②
4. $\beta(c)$ Simp ②
5. $\exists x\, \alpha(x)$ EG ③
6. $\exists x\, \beta(x)$ EG ④
7. $\exists x\, \alpha(x) \wedge \exists x\, \beta(x)$ Conj ⑤ + ⑥

Prove that if $\underbrace{\exists x(\alpha(x) \Rightarrow \beta(x)) \wedge \forall x\, \alpha(x)}_{\text{Hyp.}}$ then $\underbrace{\exists x\, \beta(x)}_{\text{Conc.}}$.

1. $\exists x(\alpha(x) \Rightarrow \beta(x)) \wedge \forall x\, \alpha(x)$ Hyp.
2. $\exists x(\alpha(x) \Rightarrow \beta(x))$ Simp ①
3. $\forall x\, \alpha(x)$ Simp ①
4. $\alpha(c) \Rightarrow \beta(c)$ EE ② c: constant
5. $\alpha(c)$ UI ③ (reusing constant c)
6. $\beta(c)$ MP ④ + ⑤
7. $\exists x\, \beta(x)$ EG ⑥

1. $\exists x\, \beta(x)$

2. $\exists x\, \alpha(x)$

3. $\beta(c)$ EI ① c: constant

~~4. $\alpha(c)$ EI ②~~ WRONG

4. $\alpha(k)$ EI ② k: constant

Given: $\forall x\,(P(x) \Rightarrow Q(x))$

$\exists x\,(R(x) \wedge \forall y\,(Q(y) \Rightarrow \sim S(x,y)))$

Prove: $\exists x\,(R(x) \wedge \forall y\,(P(y) \Rightarrow \sim S(x,y)))$